Simple Linear Regression - Marketing ROI Analysis

```
# Importing Packages
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.formula.api import ols
import statsmodels.api as sm
```

```
# Loading dataset
df = pd.read_csv('Marketing ROI.csv')
```

Data Exploration and Cleaning

```
# Rows and columns
df.shape
```

```
(4572, 4)
```

```
# Data information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   TV            4562 non-null   float64
 1   Radio         4568 non-null   float64
 2   Social_Media  4566 non-null   float64
 3   Sales         4566 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
```

```
# Total duplicate records
df.duplicated().sum()
```

```
np.int64(0)
```

```
# Number of missing values
df.isna().sum()
```

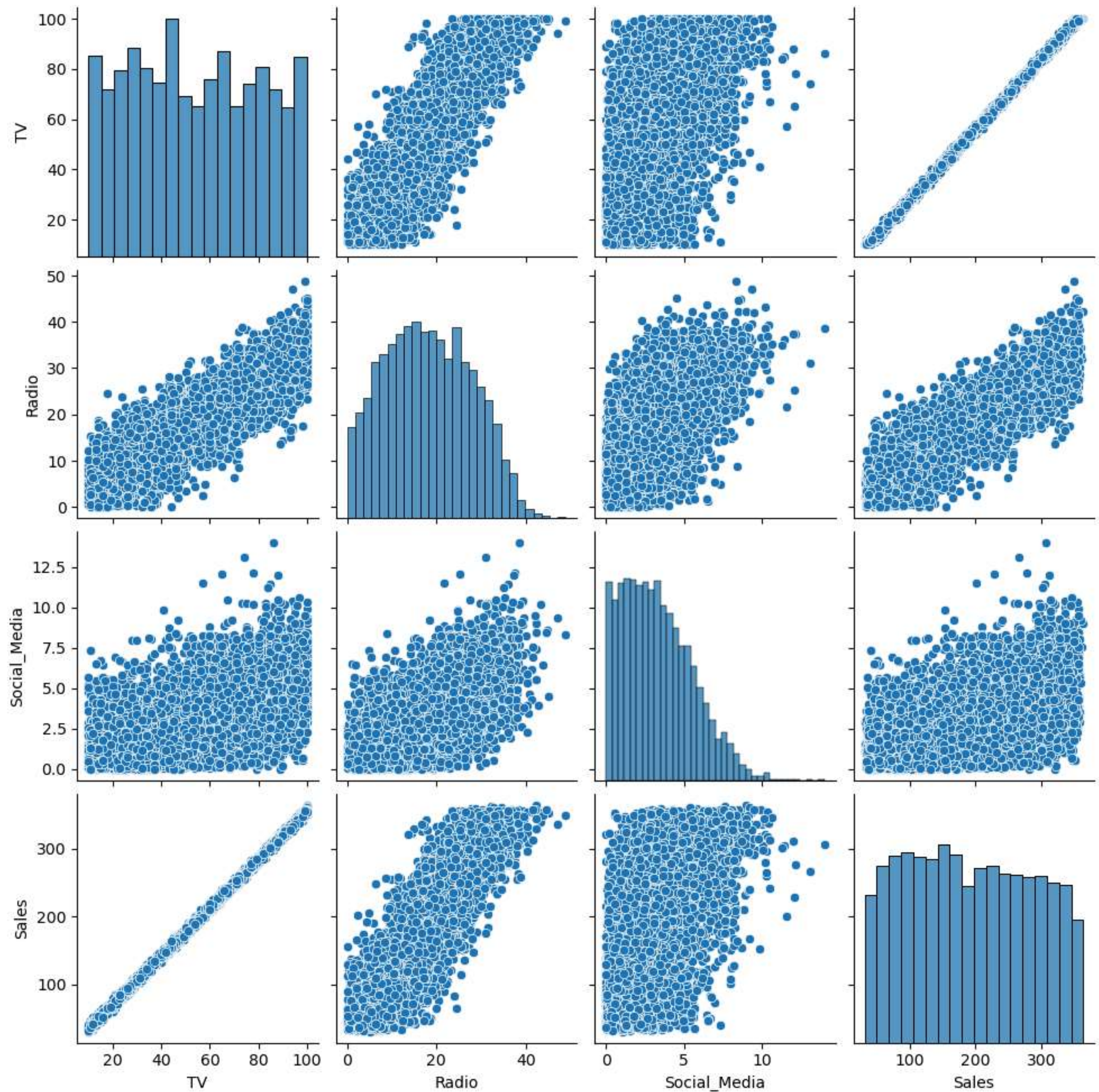
```
TV              10
Radio            4
Social_Media     6
Sales            6
dtype: int64
```

```
# Handling missing values - By removing them
df = df.dropna()
df.reset_index(inplace=True, drop=True)
```

```
# Summary statistics of numeric variables
df.describe()
```

	TV	Radio	Social_Media	Sales
count	4546.000000	4546.000000	4546.000000	4546.000000
mean	54.062912	18.157533	3.323473	192.413332
std	26.104942	9.663260	2.211254	93.019873
min	10.000000	0.000684	0.000031	31.199409
25%	32.000000	10.555355	1.530822	112.434612
50%	53.000000	17.859513	3.055565	188.963678
75%	77.000000	25.640603	4.804919	272.324236
max	100.000000	48.871161	13.981662	364.079751

```
# Pairplot diagram
sns.pairplot(df)
plt.show()
```



From the scatter plot matrix, it can be observed that there is a linear relationship between;

- TV and Sales
- Radio and Sales
- Radio and TV

However, TV and Sales had a more perfectly linear relationship

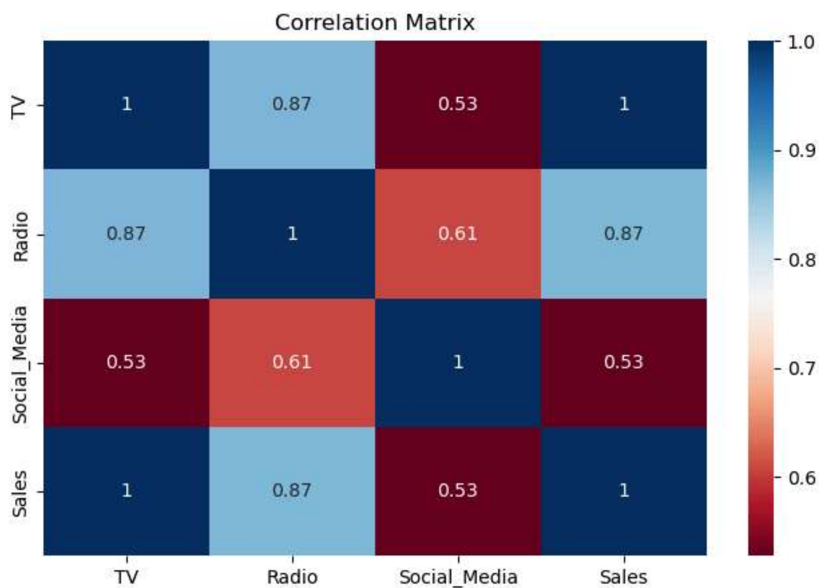
```
# Correlation between variables
corr = df.corr()
corr
```

	TV	Radio	Social_Media	Sales
TV	1.000000	0.869158	0.527687	0.999497
Radio	0.869158	1.000000	0.606338	0.868638
Social_Media	0.527687	0.606338	1.000000	0.527446
Sales	0.999497	0.868638	0.527446	1.000000

TV had the highest correlation with sales in comparison to other independent variables (Radio and Social Meadia). Hence, TV is a better predictor for sales, and should be used to build the Simple regression model

```
# Visualization of correlations
plt.figure(figsize=(8,5))
sns.heatmap(corr, cmap='RdBu', annot=True)

plt.title("Correlation Matrix")
plt.show()
```



Building the Simple Linear Regression Model

```
# Model Data
ols_data = df[["Sales", "TV", ]]
```

```
# Formula
# y = bx + c + e
ols_formula = "Sales ~ TV"
```

```
# Fitting the model
OLS = ols(ols_formula, ols_data)
model = OLS.fit()
```

Results

```
model.summary()
```

OLS Regression Results

Dep. Variable:	Sales	**R-squared:**	0.999
Model:	OLS	**Adj. R-squared:**	0.999
Method:	Least Squares	**F-statistic:**	4.517e+06
Date:	Sun, 14 Jun 2026	**Prob (F-statistic):**	0.00
Time:	10:53:53	**Log-Likelihood:**	-11366.
No. Observations:	4546	**AIC:**	2.274e+04
Df Residuals:	4544	**BIC:**	2.275e+04
Df Model:	1		
Covariance Type:	nonrobust		

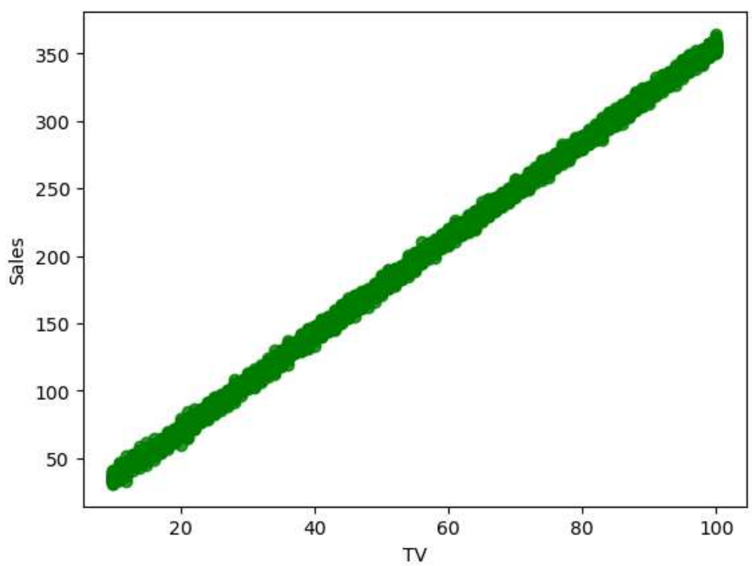
	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
Intercept	-0.1325	0.101	-1.317	0.188	-0.330	0.065
TV	3.5615	0.002	2125.272	0.000	3.558	3.565

Omnibus:	0.052	**Durbin-Watson:**	1.998
Prob(Omnibus):	0.974	**Jarque-Bera (JB):**	0.031
Skew:	-0.001	**Prob(JB):**	0.985
Kurtosis:	3.012	**Cond. No.**	138.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Checking Assumptions

```
# Linearity check
sns.regplot(x = "TV", y = "Sales", data = ols_data, color='green')
plt.show()
```




```
# Predictor variable
X = ols_data["TV"]

X.head()
```

```
0    16.0
1    13.0
2    41.0
3    83.0
4    15.0
Name: TV, dtype: float64
```

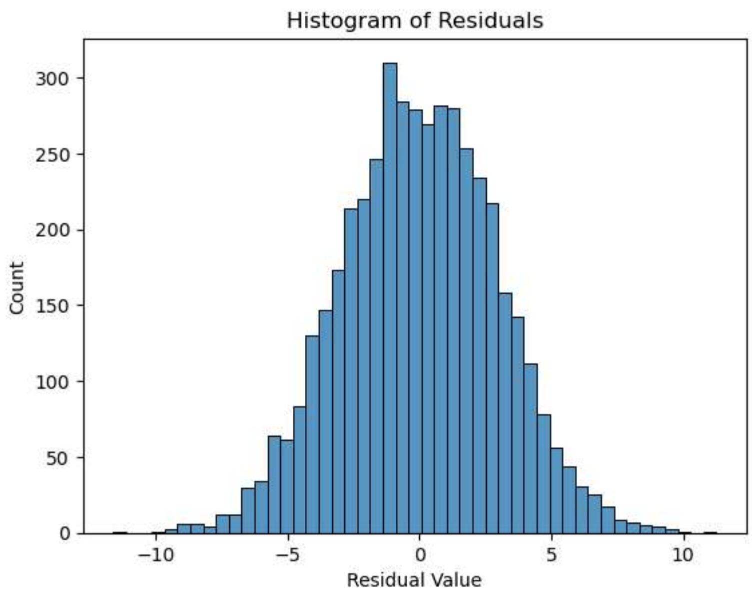
```
# Get predictions from model
fitted_values = model.predict(X)
fitted_values.head()
```

```
0     56.851733
1     46.167191
```

```
2    145.889585
3    295.473177
4     53.290219
dtype: float64
```

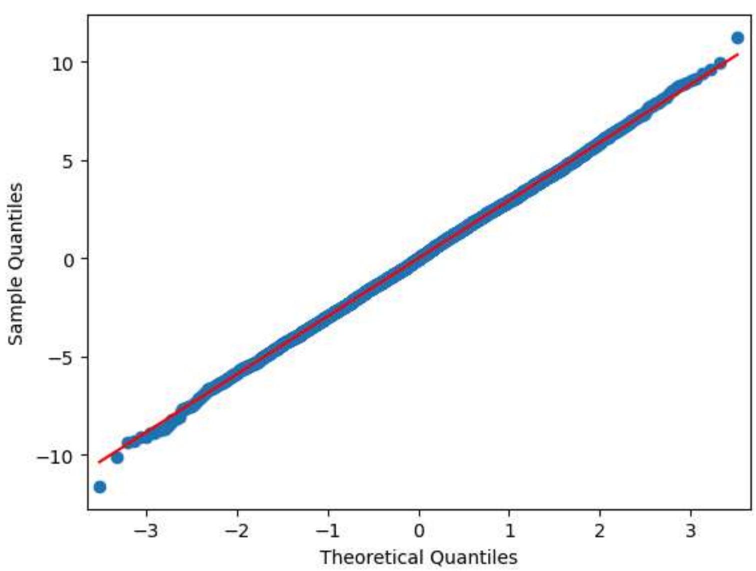
```
# Calculate residuals
residuals = model.resid
```

```
# Checking Normality Assumption
fig = sns.histplot(residuals)
fig.set_xlabel("Residual Value")
fig.set_title("Histogram of Residuals")
plt.show()
```



From the histogram above, it can be deduced that the normality assumption was met since the Residual values are almost evenly distributed

```
fig = sm.qqplot(model.resid, line = 's')
plt.show()
```

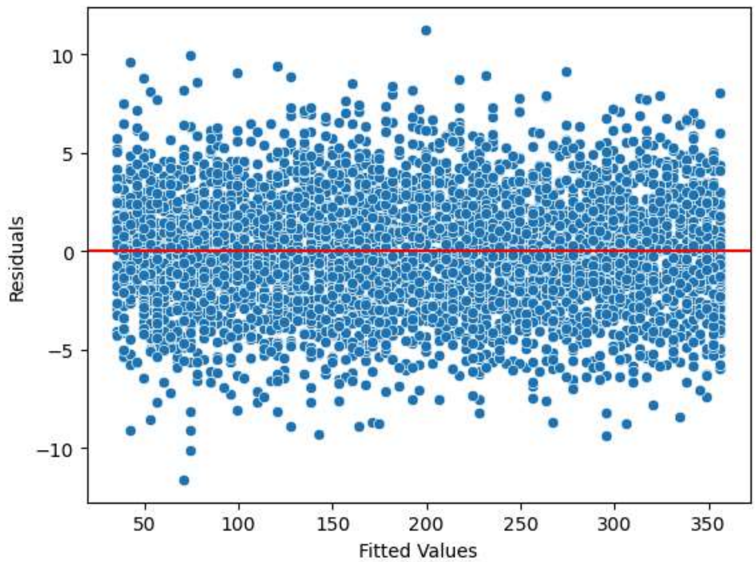


The QQ Plot above validates the findings from the histogram that the normality assumption was indeed met

```
# Homoscedasticity assumption check
fig = sns.scatterplot(x=fitted_values, y=residuals)
```

```
fig.axhline(y=0,color='red')

fig.set_xlabel("Fitted Values")
fig.set_ylabel("Residuals")
```



Key Model Results & Interpretations

- R-squared = 0.999
- TV coefficient = 3.5615
- TV p-value = 0.000
- Intercept p-value = 0.188

- The high R-squared result indicates that TV advertising investment has a very strong relationship with sales performance in this dataset and that changes in TV advertising expenditure are highly predictive of changes in sales.
- The TV coefficient value shows that for every one-unit increase in TV advertising spending, sales are expected to increase by approximately 3.56 units. This suggests that increasing investment in TV advertising is associated with a substantial increase in expected sales.
- The relationship between TV advertising and sales is statistically significant ($p < 0.001$). herefore, TV advertising is a meaningful predictor of sales and the observed relationship is unlikely to have occurred by chance.
- The intercept had a p-value of 0.188, meaning that the baseline sales value when TV advertising is zero was not statistically significant. However, the main business focus is the TV advertising coefficient, which demonstrates a significant impact on sales.

Recommendation

- Based on the regression analysis, TV advertising should receive the highest priority in future marketing budget allocation
- The organization should consider increasing or maintaining investment in TV advertising as the primary marketing channel to maximize sales returns.